

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1388

Fort Hall 100 kV
January 03, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1388. The Interconnection Customer proposes a 28.5 MW battery energy storage facility interconnecting at the Fort Hall 100 kV substation in the Ohio area. The request was received on January 16, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 534 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$987,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Japanese Mesa - Dower 100 KV Line 1
Contingency: Japanese Mesa - Dower 100 KV Ckt 2
Loading: 108.7% (Rating: 168 MVA)
DFAX: 6.8%
Pre-project loading: 107.6%
2. Facility: Gould - Dower 100 KV Line 2
Contingency: Gould - Dower 100 KV Line 1
Loading: 102.7% (Rating: 195 MVA)
DFAX: 6.7%
Pre-project loading: 101.7%
3. Facility: Ft Humbug - W Campus 345 KV
Contingency: Ft Humbug - Orchard Rd 500 KV
Loading: 101.3% (Rating: 3,061 MVA)
DFAX: 7.3%
Pre-project loading: 101.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Japanese Mesa - Dower 100 KV Line 1	Reconductor line	\$466,000	100% (\$466,000)
Gould - Dower 100 KV Line 2	Reconductor line	\$428,000	100% (\$428,000)
Ft Humbug - W Campus 345 KV	Reconductor line	\$19,480,000	0% (\$93,000)
	Total allocated to project		\$987,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.